Oral script: the 10-minute defence

Museum Guide project, MSA-DATB09-03. One section per slide, to read aloud.

How to use it

- About 1203 words: at a calm pace (about 130 words per minute), this is about 9 min 15 s. Without the 60 optional words, about 8 min 48 s. The rest of the 10 minutes is a safety margin for pauses, clicks and nerves.
- The story: a classical baseline first, then each model has to beat it (our MLP, ResNet18, a much bigger CLIP), and a last table compares them all, criterion by criterion.
- Every slide ends with a green TRANSITION box, always built the same way: "We saw that... So now...". Say it while you click to the next slide, and never skip it, even when you are late: it turns the six steps into one story.
- Read it aloud 3 times, with the slides, and time yourself once. Speak slowly; pause after each bold sentence: it is the one the jury must remember.
- Blue text in brackets is a stage cue (where to point); do not read it. Grey sentences marked (optional) can be skipped if you are late.
- If you present with a teammate, a natural split is slides 1 to 6 (data, baseline, MLP), then slides 7 to 11 (ResNet18, CLIP, recap, choice).

The story, transition by transition

1 to 2: To answer it, we did not jump straight to a big model. So now, here is our path: six steps.

2 to 3: So now, let's start with step one: the data, and how we made the test fair.

3 to 4: Our data is ready and our test is fair. So now, step two: the baseline, the score every model must beat.

4 to 5: We saw that the baseline reaches 63.5 percent. So now, step three: can our own neural network beat it?

5 to 6: We saw that our MLP is overfitting. So now, we look inside its training and try to fix it.

6 to 7: Our MLP stays below the baseline. So now, step four: a model that sees shapes, and that is fast.

7 to 8: We saw how ResNet18 works. So now, let's see what it gives on our paintings.

8 to 9: We saw that ResNet18 beats the baseline by far. So now, step five: would a much bigger model do even better?

9 to 10: We saw that CLIP is more accurate, but does not fit on a kiosk. So now, step six: all our models side by side, criterion by criterion.

Slide 1 · Is a neural network worth it on a museum kiosk?

0:00 to 0:40 · 40 s · 85 words

Good morning. Our museum wants kiosks that recognise the style of a painting from a photo.

We worked on five styles: Baroque, Cubism, Impressionism, Pop Art and Ukiyo-e. *[point at the five paintings]*

The kiosks have small CPUs, the team is small and the budget is limited. So the director asks one question: **is a neural network worth it in production, or would a simpler classical model do the job?**

> **TRANSITION TO SLIDE 2**
> To answer it, we did not jump straight to a big model. **So now, here is our path: six steps.**

Slide 2 · Six steps to answer the director

0:40 to 1:15 · 35 s · 74 words

[point at the six steps, one by one] Step one: the data, and a fair test. Step two: a baseline, a classical model without any neural network. Step three: our own neural network, an MLP. Step four: a pretrained ResNet18. Step five: a much bigger model, CLIP. Step six: all of them, criterion by criterion.

Each model must beat the baseline, and fit on a kiosk.

> **TRANSITION TO SLIDE 3**
> **So now, let's start with step one: the data, and how we made the test fair.**

Slide 3 · A fair test : one painter, one part

1:15 to 2:05 · 50 s · 106 words

We took 1 500 paintings from WikiArt, 300 per style.

We cut them into three piles: 1 061 to train, 219 to validate, that is, to choose our settings, and 220 to test, only once, at the end.

One rule matters a lot: all the paintings of a known painter go to the same pile. *[point at the zero]* With a random split, 23 of our 32 known painters would be in two piles, and a model could learn to recognise Monet instead of Impressionism. **With our split: zero.**

> **TRANSITION TO SLIDE 4**
> Our data is ready and our test is fair. **So now, step two: the baseline, the score every model must beat.**

Slide 4 · Our baseline : a classical random forest

2:05 to 2:55 · 50 s · 104 words

Our baseline uses no neural network. It cannot read raw pixels well, so we first describe each painting with 102 hand-made numbers: *[point at the second box]* 82 for its colours, and 20 for the direction and strength of its brush strokes.

[point at the tree] Then a random forest: 500 decision trees. Each tree asks yes or no questions on these numbers, like "many dark colours?", until it names a style. They vote, and the vote cancels many mistakes.

[point at the orange card] We chose its settings on validation: 63.5 percent. **This is the score to beat.**

> **TRANSITION TO SLIDE 5**
> We saw that the baseline reaches 63.5 percent. **So now, step three: can our own neural network beat it?**

Slide 5 · Our MLP : bad to generalize

2:55 to 3:50 · 55 s · 118 words

Our first neural network is the simplest one of the course: a multi-layer perceptron built from nn.Linear layers and ReLU.

Before trusting it, we ran two checks. First, a model that knows nothing should have a loss of ln 5, about 1.61. *(optional) Ours started at 1.99, too confident, so we made its last layer smaller and got 1.59.* Second, it must be able to learn one small batch by heart: it does, 100 percent.

[point at the curves] The training accuracy goes to 99.7 percent, but the validation accuracy stays around 40 percent. **This is overfitting: the network learns the paintings by heart.**

> **TRANSITION TO SLIDE 6**
> We saw that our MLP is overfitting. **So now, we look inside its training and try to fix it.**

Slide 6 · Healthy training, but still below the baseline

3:50 to 4:40 · 50 s · 109 words

Before fixing it, we looked inside the network: the gradient reaches every layer and only 4 percent of the neurons are dead. The training is healthy: the problem really is overfitting.

Then we attacked it with the tools of session 6. *[point at the chart]* Together, weight decay, dropout, batch norm, flips and early stopping take the MLP from 38.8 to 45.5 percent, and the gap from 0.61 to 0.12.

[point at the orange bar] But it stays far below the baseline: 45.5 against 63.5. **It finds only 7 percent of the Cubist paintings: it sees colours, not shapes.**

> **TRANSITION TO SLIDE 7**
> Our MLP stays below the baseline. **So now, step four: a model that sees shapes, and that is fast.**

Slide 7 · ResNet18: a network made for images

4:40 to 6:05 · 85 s · 187 words

To see shapes, we need a network made for images: a convolutional network, or CNN. The brief asks for a ResNet, a family of CNNs that won the ImageNet competition in 2015. ResNet18 is its smallest member: light enough for a kiosk, and PyTorch gives it already trained.

[point at the dark box] Its 18 layers slide small filters over the picture: the first layers find edges and colours, the next ones textures, the last ones shapes. *(optional) "Res" means residual: shortcuts between its layers let a deep network learn well.*

ResNet18 already learned this on 1.2 million ImageNet photos, and we only have 1 061 paintings, so we never retrain all of it. *[point at the 512 numbers]* It turns each painting into 512 numbers, and we replace its last layer by our own, for our 5 styles.

[point at the bottom right card] We tried two ways. Frozen: only our new layer learns, 2 565 weights. Fine-tuned: the last block learns again too, 8.4 million weights. **Like hiring an experienced art expert: frozen, we only teach them our five style names; fine-tuned, we also retrain their eye.**

> **TRANSITION TO SLIDE 8**
> We saw how ResNet18 works. **So now, let's see what it gives on our paintings.**

Slide 8 · A pretrained ResNet18 sees shapes

6:05 to 7:10 · 65 s · 146 words

[point at the top bar] These are the final tests, run once at the very end: 440 paintings never seen before, our test pile and 220 new paintings from another part of WikiArt. The fine-tuned ResNet18 gets 87.8 percent: **22 points above the baseline, and twice our MLP.**

[point at the middle card] On validation, the frozen and the fine-tuned versions were too close to choose, on only 219 paintings. On the tests, the fine-tuned one wins by 5 points, and it is not luck. **That is why we keep the fine-tuned one.**

[point at the top card] And it does see shapes: on validation, the ResNet finds 71 percent of the Cubist paintings, against 7 for our MLP.

[point at the bottom card] (optional) We monitored it like the MLP: the gap between train and validation is only 0.09, against 0.61 for our first MLP.

> **TRANSITION TO SLIDE 9**
> We saw that ResNet18 beats the baseline by far. **So now, step five: would a much bigger model do even better?**

Slide 9 · CLIP ViT-L/14: more accurate, but too big

7:10 to 8:15 · 65 s · 139 words

ResNet18 is one of the smallest image networks, so we also tried one of the biggest: CLIP ViT-L/14, by OpenAI. A transformer, like BERT, that learned from 400 million Internet images and their captions. We used it like ResNet18: frozen, with our linear head.

[point at the big number] On validation, it reaches 96.3 percent: **the best of all our models.**

[point at the three rows] But look at the price. One photo takes 4.6 seconds on one CPU thread, 38 times ResNet18, and its file weighs 1.2 gigabytes, 27 times more. **On a kiosk, that is about 18 seconds per photo: too slow and too big. It would need a GPU.**

(optional) And it has probably seen WikiArt paintings during its training.

> **TRANSITION TO SLIDE 10**
> We saw that CLIP is more accurate, but does not fit on a kiosk. **So now, step six: all our models side by side, criterion by criterion.**

Slide 10 · Every model, by criteria: ResNet18 stands out

8:15 to 9:15 · 60 s · 135 words

[point at the accuracy column] On the same 440 new paintings, the baseline gets 65 percent, our MLP 42, the fine-tuned ResNet18 88, and CLIP 96.

[point at the other columns] Then the kiosk criteria, fixed before the tests: one second per photo on the kiosk, and a file under 200 megabytes. *[point at the blue row]* ResNet18 answers in 121 milliseconds on our laptop, weighs 45 megabytes and retrains in 3.4 minutes. *[point at the CLIP row]* CLIP breaks both limits.

ResNet18 is the only model that is both accurate and fits the kiosk.

[point at the blue banner] **So our final choice is a fine-tuned ResNet18: it answers the director's question. Yes, a neural network is worth it.**

One visitor in eight gets a wrong style, against one in three with the baseline, and it runs on a modest kiosk. Before launch, we test it on real kiosk photos.

Thank you. We are happy to take your questions.